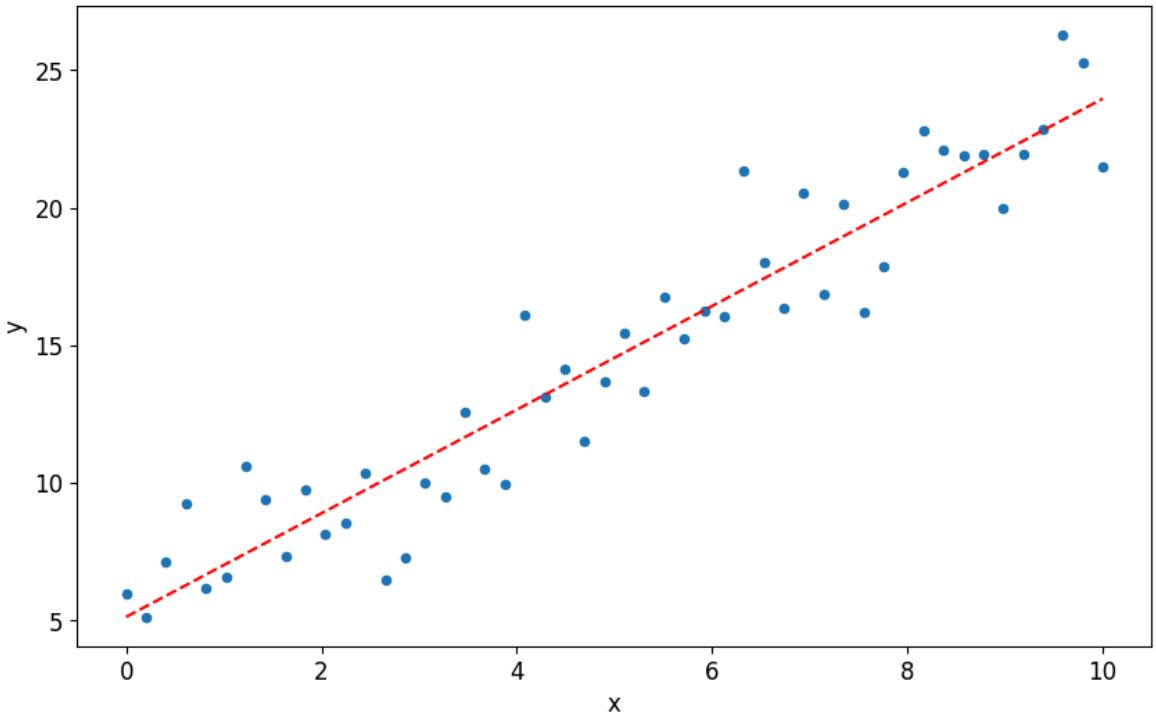
In [1]:
```
import numpy as np
import matplotlib.pyplot as plt
import pandas as pd
from scipy import stats
from scipy.integrate import quad
import seaborn as sns
from scipy import stats


plt.rcParams.update({'font.size': 12})
plt.rcParams['figure.figsize'] = (10, 6)
```

In [ ]:
```
# Ustawienie seed dla powtarzalności
np.random.seed(42)

# Generowanie danych: y ≈ 2x + 5 + szum
n = 50
x = np.linspace(0, 10, n)
y_true = 2 * x + 5  # prawdziwa zależność
noise = np.random.normal(0, 2, n)  # losowy szum
y = y_true + noise

# Stworzenie DataFrame
data = pd.DataFrame({'x': x, 'y': y})


slope, intercept, r_value, p_value, std_err = stats.linregress(data.x, data.y)
print(f"slope: {slope:.2f}, intercept: {intercept:.2f}")
fig, ax = plt.subplots()
data.plot(kind="scatter", x='x', y='y', ax=ax)
y_linreg = slope*x + intercept
ax.plot(x, y_linreg, 'r--')
```

```
slope: 1.88, intercept: 5.13
```

Out[ ]:
```
[<matplotlib.lines.Line2D at 0x79b36a3780e0>]
```



In [ ]:
```
def fit_and_plot(x, y, ax):
  slope, intercept, r_value, p_value, std_err = stats.linregress(x, y)
  print(f"slope: {slope:.2f}, intercept: {intercept:.2f}")
  print(f"r: {r_value}")
  print(f"err: {std_err}")
  ax.plot(x, y, 'bo')
  x_linspace = np.linspace(min(x), max(x), 30)
  y_linreg = slope*x_linspace + intercept
  ax.plot(x_linspace, y_linreg, 'r-')
```

In [ ]:
```
np.savetxt("dane_do_fit.txt", np.c_[x, y])
```

In [ ]:
```
df = pd.read_csv("dane_do_fit.txt", delimiter=" ", header=None, names=["x", "y"]
```

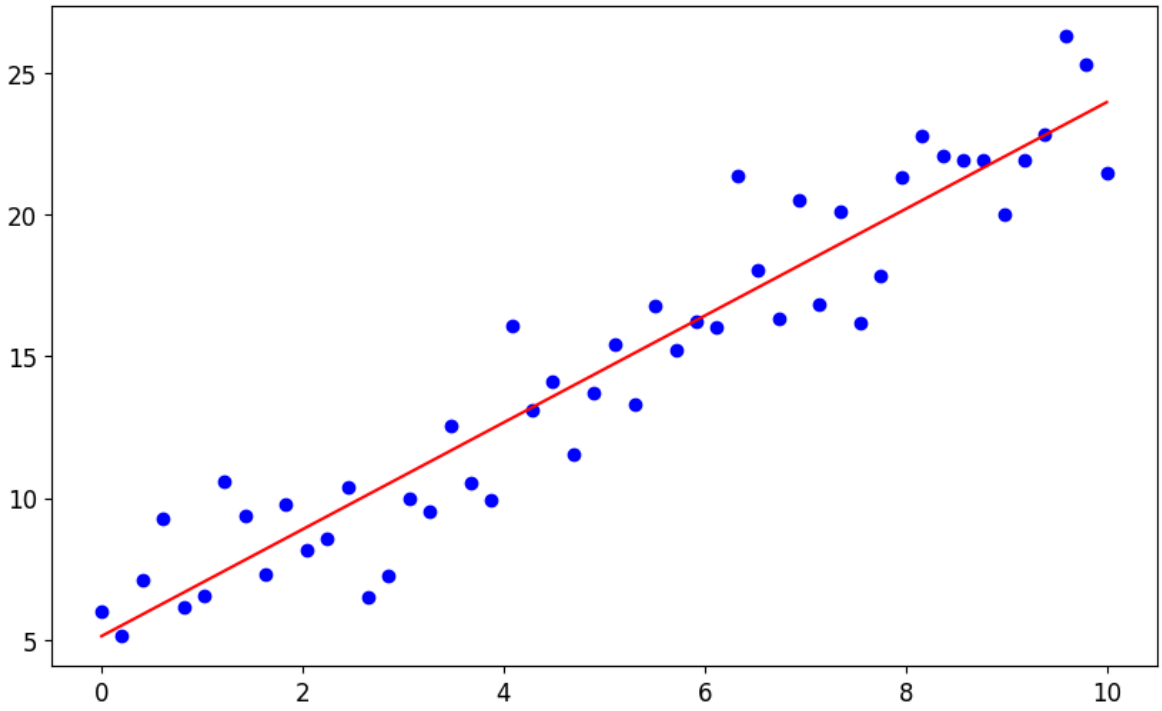
In [ ]:
```
df.head()
```

Out[ ]:

	x	y
0	0.000000	5.993428
1	0.204082	5.131635
2	0.408163	7.111704
3	0.612245	9.270550
4	0.816327	6.164346

In [ ]:
```
fig, ax = plt.subplots()
fit_and_plot(df['x'], df['y'], ax)
```

```
slope: 1.88, intercept: 5.13
r: 0.9503548762459013
err: 0.08903837899142658
```



Zadanie 1

1. **Wczytaj dane:**
   - Wczytaj dane z pliku `anscombe.csv` do DataFrame za pomocą biblioteki `pandas`.
   - Przekształć dane, aby miały format:

```
    x1    y1    x2    y2    x3     y3   x4    y4
0  10.0  8.04  10.0  9.14  10.0   7.46  8.0  6.58
1   8.0  6.95   8.0  8.14   8.0   6.77  8.0  5.76
```

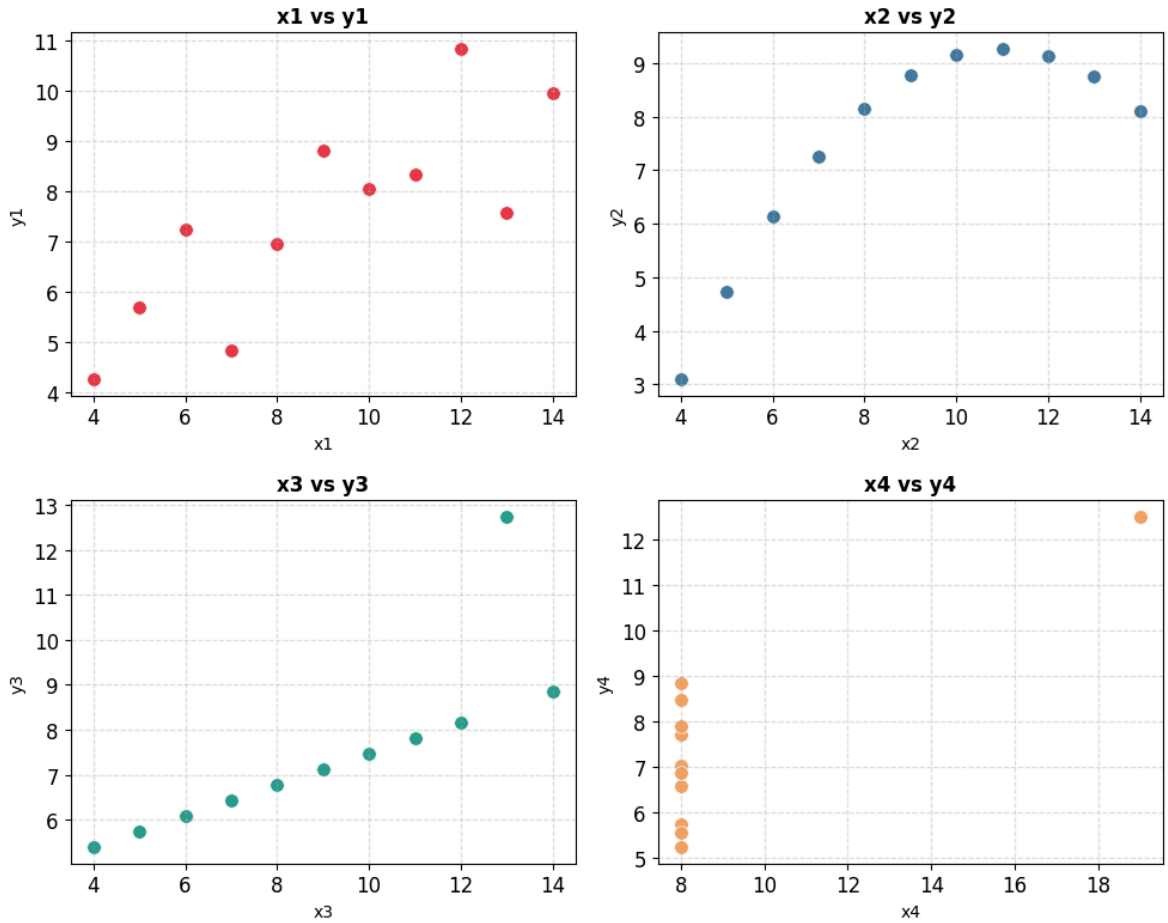
2. **Stwórz wykresy punktowe:**
   - Użyj biblioteki `matplotlib` lub `seaborn`, aby stworzyć wykresy.
   - Na każdej osi (`ax`) umieść wykres punktowy dla par `x1, y1`, `x2, y2`, `x3, y3` oraz `x4, y4`.

In [ ]:
```
! head anscombe.csv
```

```
0,0,1,1,2,2,3,3
x,y,x,y,x,y,x,y
10.0,8.04,10.0,9.14,10.0,7.46,8.0,6.58
8.0,6.95,8.0,8.14,8.0,6.77,8.0,5.76
13.0,7.58,13.0,8.74,13.0,12.74,8.0,7.71
9.0,8.81,9.0,8.77,9.0,7.11,8.0,8.84
11.0,8.33,11.0,9.26,11.0,7.81,8.0,8.47
14.0,9.96,14.0,8.10,14.0,8.84,8.0,7.04
6.0,7.24,6.0,6.13,6.0,6.08,8.0,5.25
4.0,4.26,4.0,3.10,4.0,5.39,19.0,12.50
```

In [3]:
```
raw = pd.read_csv("anscombe.csv", header=None)

columns = ["x1", "y1", "x2", "y2", "x3", "y3", "x4", "y4"]
df = raw.iloc[2:].reset_index(drop=True)
df.columns = columns
df = df.astype(float)

print(df.head(2).to_string())

fig, axes = plt.subplots(2, 2, figsize=(10, 8))
pairs = [("x1", "y1"), ("x2", "y2"), ("x3", "y3"), ("x4", "y4")]
colors = ["#E63946", "#457B9D", "#2A9D8F", "#F4A261"]

for ax, (x, y), color in zip(axes.flat, pairs, colors):
    ax.scatter(df[x], df[y], color=color, s=70, edgecolors="white", linewidths=0
    ax.set_title(f"{x} vs {y}", fontsize=12, fontweight="bold")
    ax.set_xlabel(x, fontsize=10)
    ax.set_ylabel(y, fontsize=10)
    ax.grid(True, linestyle="--", alpha=0.4)

plt.tight_layout()
plt.show()
```

```
    x1    y1    x2    y2    x3    y3   x4    y4
0  10.0  8.04  10.0  9.14  10.0  7.46  8.0  6.58
1   8.0  6.95   8.0  8.14   8.0  6.77  8.0  5.76
```

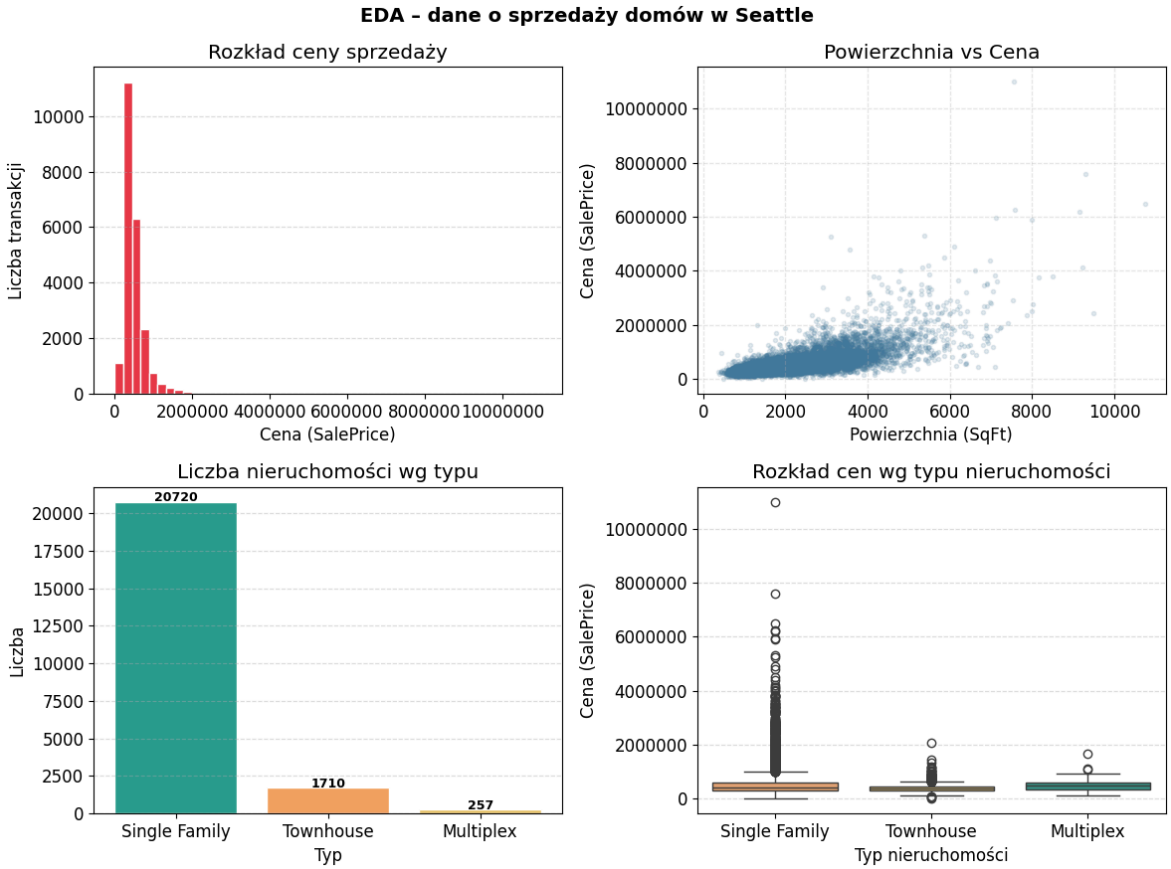


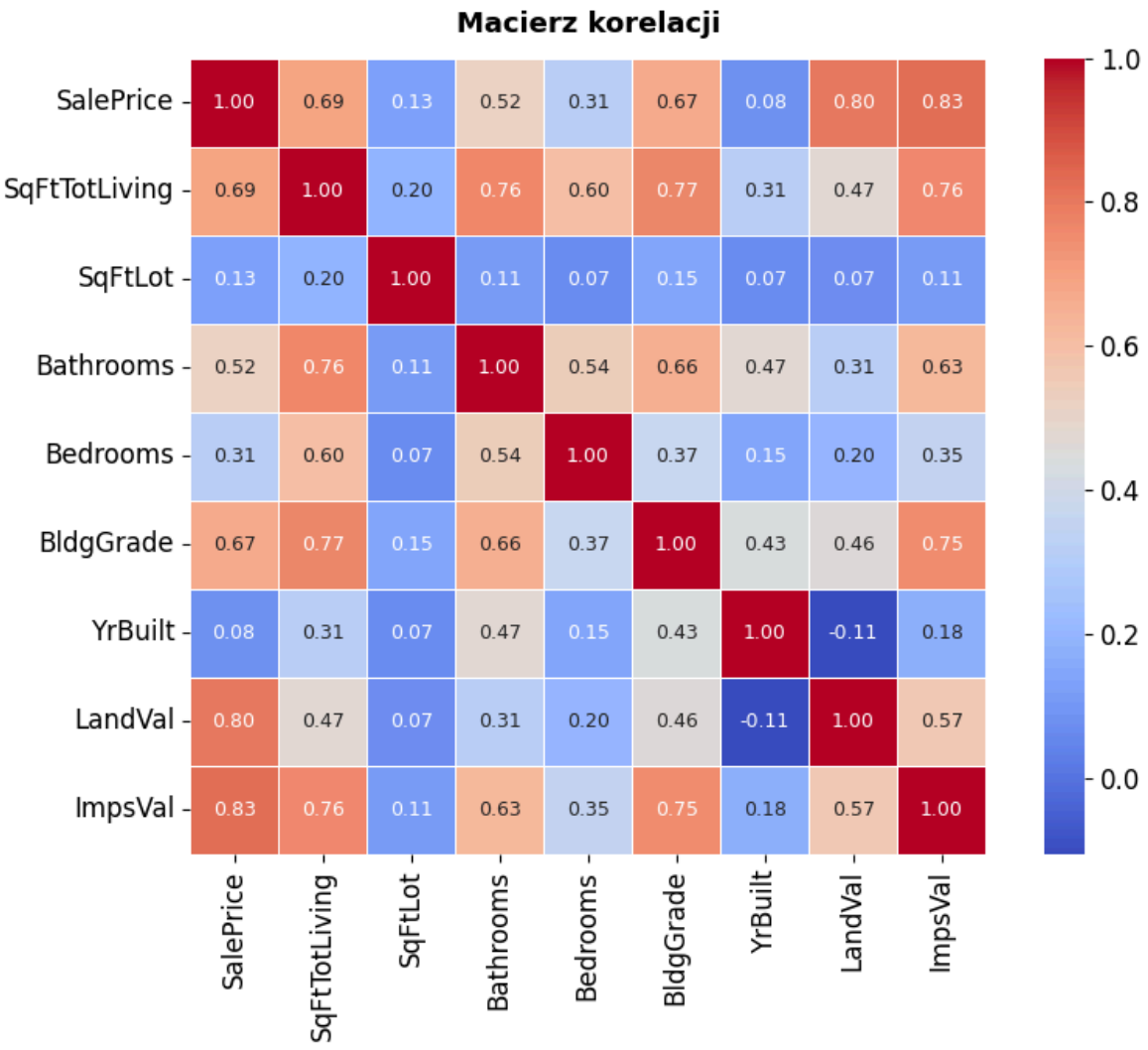
Zadanie 2

Przeprowadź eksploracyjną analizę danych (EDA) dla zbioru danych o domach `house_train.csv`

In [8]:
```
df = pd.read_csv("house_sales.csv", sep="\t")
df["DocumentDate"] = pd.to_datetime(df["DocumentDate"])
df["Year"] = df["DocumentDate"].dt.year

fig, axes = plt.subplots(2, 2, figsize=(12, 9))
fig.suptitle("EDA – dane o sprzedaży domów w Seattle", fontsize=14, fontweight="

ax = axes[0, 0]
ax.hist(df["SalePrice"], bins=50, color="#E63946", edgecolor="white")
ax.set_title("Rozkład ceny sprzedaży")
ax.set_xlabel("Cena (SalePrice)")
ax.set_ylabel("Liczba transakcji")
ax.ticklabel_format(style="plain", axis="x")
ax.grid(axis="y", linestyle="--", alpha=0.4)

ax = axes[0, 1]
ax.scatter(df["SqFtTotLiving"], df["SalePrice"], alpha=0.15, s=8, color="#457B9D
ax.set_title("Powierzchnia vs Cena")
ax.set_xlabel("Powierzchnia (SqFt)")
ax.set_ylabel("Cena (SalePrice)")
ax.ticklabel_format(style="plain", axis="y")
ax.grid(linestyle="--", alpha=0.3)

ax = axes[1, 0]
```

```
counts = df["PropertyType"].value_counts()
ax.bar(counts.index, counts.values, color=["#2A9D8F", "#F4A261", "#E9C46A"], edg
for i, v in enumerate(counts.values):
    ax.text(i, v + 50, str(v), ha="center", fontsize=9, fontweight="bold")
ax.set_title("Liczba nieruchomości wg typu")
ax.set_xlabel("Typ")
ax.set_ylabel("Liczba")
ax.grid(axis="y", linestyle="--", alpha=0.4)

ax = axes[1, 1]
order = df["PropertyType"].value_counts().index.tolist()
sns.boxplot(data=df, x="PropertyType", y="SalePrice", order=order,
            hue="PropertyType", palette=["#2A9D8F", "#F4A261", "#E9C46A"],
            legend=False, ax=ax)
ax.set_title("Rozkład cen wg typu nieruchomości")
ax.set_xlabel("Typ nieruchomości")
ax.set_ylabel("Cena (SalePrice)")
ax.ticklabel_format(style="plain", axis="y")
ax.grid(axis="y", linestyle="--", alpha=0.4)

plt.tight_layout()
plt.show()

num_cols = ["SalePrice", "SqFtTotLiving", "SqFtLot", "Bathrooms",
            "Bedrooms", "BldgGrade", "YrBuilt", "LandVal", "ImpsVal"]
corr = df[num_cols].corr()

fig2, ax2 = plt.subplots(figsize=(9, 7))
sns.heatmap(corr, annot=True, fmt=".2f", cmap="coolwarm",
            square=True, linewidths=0.5, ax=ax2, annot_kws={"size": 9})
ax2.set_title("Macierz korelacji", fontsize=13, fontweight="bold", pad=12)
plt.tight_layout()
plt.show()
```




Macierz korelacji
SalePrice 1.00 0.69 0.13 0.52 0.31 0.67 0.08 0.80 0.83
SqFtTotLiving 0.69 1.00 0.20 0.76 0.60 0.77 0.31 0.47 0.76
SqFtLot 0.13 0.20 1.00 0.11 0.07 0.15 0.07 0.07 0.11
Bathrooms 0.52 0.76 0.11 1.00 0.54 0.66 0.47 0.31 0.63
Bedrooms 0.31 0.60 0.07 0.54 1.00 0.37 0.15 0.20 0.35
BldgGrade 0.67 0.77 0.15 0.66 0.37 1.00 0.43 0.46 0.75
YrBuilt 0.08 0.31 0.07 0.47 0.15 0.43 1.00 -0.11 0.18
LandVal 0.80 0.47 0.07 0.31 0.20 0.46 -0.11 1.00 0.57
ImpsVal 0.83 0.76 0.11 0.63 0.35 0.75 0.18 0.57 1.00
SalePrice
SqFtTotLiving
SqFtLot
Bathrooms
Bedrooms
BldgGrade
YrBuilt
LandVal
ImpsVal
1.0
0.8
0.6
0.4
0.2
0.0